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- Practice problems

McMaster University

Mathematics and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03
Introduction to Differential Equations

Instructor: David Earn

Lecture 15
Higher-order linear ODE theory
Friday 9 October 2026

Announcements

- The TA is Illya Kierkosz (`kierkosi@mcmaster.ca`)
  - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: `https://davidearn.github.io/math2c/`
- The course outline is on the Simple Syllabus web site.
- Avenue to Learn provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on `childsmath`.
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
  - `Subject: Math 2C03: confusion about assignment 1`
  - `Subject: Math 2C03: conflict with test window`
- Office hour: Thursdays, 9:30–10:20 am, after class.

n^{th} order linear IVPs

The most general ***n^{th} order linear ODE*** can be written

$$a_n(x)\frac{\mathrm{d}^n y}{\mathrm{d}x^n}+a_{n-1}(x)\frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}}+\cdots+a_1(x)\frac{\mathrm{d}y}{\mathrm{d}x}+a_0(x)y \;=\; g(x) \qquad (1)$$

where the $n+2$ functions $a_n(x)$, $a_{n-1}(x)$, ..., $a_0(x)$, $g(x)$, are defined for all x in an interval $\mathcal{I}$.

- The functions $a_i(x)$, for $i = 0, 1, \ldots, n$, and $g(x)$ *can be nonlinear functions of x*.
- If $g(x) \equiv 0$ then the ODE is ***homogeneous***.
- The linear ODE (1) is ***normal on $\mathcal{I}$*** if $a_0(x), \ldots, a_n(x)$ and $g(x)$ are continuous on $\mathcal{I}$, and $a_n(x) \neq 0$ throughout $\mathcal{I}$.

We obtain an ***n^{th} order Initial Value Problem (IVP)*** if we specify n initial conditions:

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad \ldots, \quad y^{(n-1)}(x_0) = y_{n-1}, \qquad (2)$$

where the n numbers $y_0, y_1, \ldots, y_{n-1} \in \mathbb{R}$ are constants.

n^{th} order linear IVPs

Theorem (Existence and Uniqueness for Linear ODEs [Trench 9.1.1])

Let $\mathcal{I} = (a, b)$ be an interval and $x_0 \in \mathcal{I}$. Suppose the ODE (1) is normal on $\mathcal{I}$. Then the IVP consisting of the ODE (1) and the initial conditions (2) has a unique solution $y(x)$ defined on all of $\mathcal{I}$.

Note: Unlike the existence-uniqueness theorem that we discussed for first order ODEs, the theorem above guarantees a unique solution on the *whole interval* $\mathcal{I}$. But this theorem applies only to *linear* ODEs.

Different initial conditions for linear ODEs

Suppose the ODE (1) is normal on $\mathcal{I}$, and $x_0 \in \mathcal{I}$.

- The standard n initial values determine $y^{(n)}(x_0)$ too: evaluate the ODE at x_0 and solve for it, since $a_n(x_0) \neq 0$.
- What if we prescribe $y^{(n)}(x_0)$ instead of one of the standard values $y^{(i)}(x_0)$, where $0 \leq i < n$ and $y^{(0)} = y$?
- Evaluate the ODE at x_0:

$$a_i(x_0)y^{(i)}(x_0) = g(x_0) - \sum_{\substack{k=0 \\ k\neq i}}^{n} a_k(x_0)y^{(k)}(x_0).$$

  If $a_i(x_0) \neq 0$, divide by it to recover the missing value and apply the existence–uniqueness theorem. Here, a_i need be nonzero only at x_0.
- If $a_i(x_0) = 0$, we cannot recover the missing value. Instead there is either no solution or infinitely many solutions.
  - Inconsistent initial conditions give *no solution*

    Example with $a_1(x) = a_0(x) \equiv 0$: $y'' = 0,\ y'(0) = 1,\ y''(0) = 1 \implies$ no solution.
  - Consistent initial conditions give *a one-parameter family*.

    Example with $a_1(x) = a_0(x) \equiv 0$:

    $y'' = 0,\ y'(0) = 1,\ y''(0) = 0 \implies y = x + C,\ C \in \mathbb{R}$.

Differential Operators

Derivative operator D:

$$Dy = \frac{\mathrm{d}y}{\mathrm{d}x}, \qquad D^2y = D(Dy) = \frac{\mathrm{d}^2y}{\mathrm{d}x^2}, \qquad \ldots, \qquad D^ny = \frac{\mathrm{d}^ny}{\mathrm{d}x^n}$$

n^{th} order linear differential operator:

$$L = a_n(x)D^n + a_{n-1}(x)D^{n-1} + \cdots + a_1(x)D + a_0(x)$$

Any such L is linear: If $y_1(x)$ and $y_2(x)$ are n-times differentiable functions and $\alpha_1, \alpha_2 \in \mathbb{R}$ then:

$$L[\alpha_1 y_1(x) + \alpha_2 y_2(x)] \;=\; \alpha_1 L[y_1(x)] + \alpha_2 L[y_2(x)]$$

The most general n^{th} order linear ODE (1) can be written

$$L[y] = g(x)$$

Superposition Principle

When is the sum of solutions to an ODE also a solution?

Answer: Whenever the ODE is *linear and homogeneous*.

Theorem (Superposition Principle [Trench §9.1])

If $y_1, y_2, \ldots, y_k$ are solutions of $L[y] = 0$ on an interval $\mathcal{I}$, then any linear combination

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x)\,,$$

is also a solution of $L[y] = 0$ on the interval $\mathcal{I}$.

Proof on next slide...

Note: The terminology "superposition" comes from wave theory.

Superposition Principle

Proof of Superposition Principle.

Suppose $L[y_i] = 0$ for each $i = 1, 2, \ldots, k$.

$$
\begin{aligned}
&L[c_1y_1 + c_2y_2 + \cdots + c_ky_k] \\
&\qquad = L[c_1y_1] + L[c_2y_2 + \cdots + c_ky_k] && L \text{ is linear} \\
&\qquad = L[c_1y_1] + L[c_2y_2] + L[c_3y_3 + \cdots + c_ky_k] && L \text{ is linear} \\
&\qquad \vdots \\
&\qquad = L[c_1y_1] + L[c_2y_2] + \cdots + L[c_ky_k] && L \text{ is linear} \\
&\qquad = c_1L[y_1] + c_2L[y_2] + \cdots + c_kL[y_k] && L \text{ is linear} \\
&\qquad = c_1 \cdot 0 + c_2 \cdot 0 + \cdots + c_k \cdot 0 && L[y_i] = 0 \\
&\qquad = 0
\end{aligned}
$$

$\Longrightarrow$ $c_1y_1 + c_2y_2 + \cdots + c_ky_k$ is a solution of $L[y] = 0$.

□

Superposition Principle

Important special cases of the superposition principle:

- For a homogeneous linear ODE, a constant multiple of a solution is a solution: $L[cy] = c\,L[y] = c \cdot 0 = 0$ for any $c \in \mathbb{R}$
- Considering $c = 0$: the trivial solution, $y(x) \equiv 0$, is always a solution to a homogeneous ODE: $L[0] = 0$.

Linear independence and dependence of functions

Definition (Linear independence/dependence)

The k functions $f_1(x), \ldots f_k(x)$ defined on the interval $\mathcal{I}$ are said to be ***linearly independent on*** $\mathcal{I}$ if whenever their linear combination vanishes on $\mathcal{I}$, *i.e.*,

$$c_1 f_1(x) + c_2 f_2(x) + \cdots + c_k f_k(x) = 0 \qquad \text{for all } x \in \mathcal{I}$$

it follows that $\quad c_1 = c_2 = \cdots = c_k = 0\,.$

- If $f_1(x), \ldots f_k(x)$ are not linearly independent on $\mathcal{I}$, they are said to be ***linearly dependent on*** $\mathcal{I}$.
- If they are linearly dependent, then at least one of the functions $f_1(x), \ldots f_k(x)$ can be expressed as a linear combination of the others (on $\mathcal{I}$).

Linear independence and dependence of functions

Example (Linear independence/dependence)

Show that the functions $\cos x$, $\sin x$, and the constant function 1, are linearly independent on $\mathcal{I} = (-\pi, \pi)$.

Solution: Suppose

$$c_1 \cos x + c_2 \sin x + c_3(1) = 0 \qquad \text{for all } x \in (-\pi, \pi).$$

For $x = 0$ we get:

$$c_1 \cos 0 + c_2 \sin 0 + c_3(1) = c_1 \cdot 1 + c_2 \cdot 0 + c_3 = c_1 + c_3 = 0 \implies c_1 = -c_3$$

whereas for $x = \frac{\pi}{2}$ we get:

$$c_1 \cos \frac{\pi}{2} + c_2 \sin \frac{\pi}{2} + c_3(1) = c_1 \cdot 0 + c_2 \cdot 1 + c_3 = c_2 + c_3 = 0 \implies c_2 = -c_3$$

...cont...

Linear independence and dependence of functions

Example (Linear independence/dependence)

and for $x = -\frac{\pi}{2}$ we get:

$$c_1 \cos -\frac{\pi}{2} + c_2 \sin -\frac{\pi}{2} + c_3(1) = c_1 \cdot 0 - c_2 \cdot 1 + c_3$$
$$= -c_2 + c_3 = 0 \implies c_2 = c_3$$

Adding $c_2 = -c_3$ and $c_2 = c_3$, we have

$$\begin{aligned} 2c_2 = 0 &\implies c_2 = 0 \\ &\implies c_3 = 0 \qquad \because c_2 = c_3 \\ &\implies c_1 = 0 \qquad \because c_1 = -c_3 \\ &\implies c_1 = c_2 = c_3 = 0 \end{aligned}$$

which implies $\cos x$, $\sin x$, and the constant function 1, are linearly independent on $\mathcal{I} = (-\pi, \pi)$.

Poll

- Go to
  https://www.childsmath.ca/childsa/forms/main_login.php
- Click on Math 2C03
- Click on Take Class Poll
- Fill in poll **Linear independence of trigonometric functions**
- Submit.

Linear independence and dependence of functions

Example (Linear independence/dependence)

Consider $\cos^2 x$, $\sin^2 x$, and the constant function 1. Are these three functions linearly independent on the interval $\mathcal{I} = (-\pi, \pi)$?

Answer: No, they are linearly dependent.

Solution: We must find constants $c_1, c_2, c_3 \in \mathbb{R}$ (*not all zero*) such that

$$c_1 \cos^2 x + c_2 \sin^2 x + c_3 = 0.$$

One such solution:

$$c_1 = 1, \qquad c_2 = 1, \qquad c_3 = -1.$$

Linear independence of solutions of linear ODEs

Distinct solutions of a linear ODE are more useful if they are linearly independent.

For a *homogeneous linear n^{th} order ODE*, if we have *n linearly independent solutions*, $y_1(x), y_2(x), \ldots, y_n(x)$, on an interval $\mathcal{I}$, then we call the set $\{y_1, y_2, \ldots, y_n\}$, a ***fundamental set of solutions*** on the interval $\mathcal{I}$.

Theorem (Existence of a Fundamental Set [Trench §9.1])

Suppose $L[y] = 0$ is normal on $\mathcal{I}$. Then there exists a fundamental set of solutions on $\mathcal{I}$.

This theorem is important because any solution of $L[y] = 0$ can be expressed as a linear combination of the fundamental set...

Linear independence of solutions of linear ODEs

Theorem (General Solution of Homogeneous n^{th} order ODE [Trench 9.1.4])

*If $\{y_1, y_2, \ldots, y_n\}$ is a fundamental set of solutions for the nth order ODE $L[y] = 0$ on an interval $\mathcal{I}$, then the **general solution** of $L[y] = 0$ on the interval $\mathcal{I}$ is*

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x)$$

where c_i, $i = 1, 2, \ldots, n$, are arbitrary constants.

- Given this theorem, it is very important to know if a set of solutions is linearly independent.
- Checking linear independence directly from the definition can be cumbersome. Fortunately, there is a convenient criterion based on a determinant that is usually easier to use.

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Mathematics 2C03
Introduction to Differential Equations

Instructor: David Earn

Lecture 16
Wronskians and general solutions of linear ODEs
Tuesday 20 October 2026

Poll

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- Fill in poll **Test 1 experience survey**
- Submit.

Linear independence of solutions of linear ODEs

Given a set of n functions $\{f_1(x), f_2(x), \ldots, f_n(x)\}$ that are at least $n-1$ times differentiable, the ***Wronskian*** of the functions is the determinant

$$W(f_1, f_2, \ldots, f_n) = \begin{vmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \cdots & f_n^{(n-1)} \end{vmatrix}$$

Example (Wronskian of two functions, x^2 and $\sin x$)

$$W(x^2, \sin x) = \begin{vmatrix} x^2 & \sin x \\ 2x & \cos x \end{vmatrix} = x^2 \cos x - 2x \sin x$$

Linear independence of solutions of linear ODEs

Theorem (Criterion for Linearly Independent Solutions [Trench 9.1.4])

Suppose $L[y] = 0$ is normal on $\mathcal{I}$. Consider a set of n solutions $\{y_1, y_2, \ldots, y_n\}$ of the nth order homogeneous linear ODE, $L[y] = 0$, on an interval $\mathcal{I}$. The set of n solutions is ***linearly independent*** *on $\mathcal{I}$* ***if and only if*** *the* ***Wronskian*** *$W(y_1, y_2, \ldots, y_n) \neq 0$ for every $x \in \mathcal{I}$.*

- If $y_1, y_2, \ldots, y_n$ are solutions of $L[y] = 0$ on the interval $\mathcal{I}$, then ***either*** $W(y_1, y_2, \ldots, y_n) \equiv 0$ on $\mathcal{I}$ (*everywhere zero*) ***or*** $W(y_1, y_2, \ldots, y_n) \neq 0$ on $\mathcal{I}$ (*nowhere zero*).
  - *Note:* This is not guaranteed for arbitrary functions $y_1, y_2, \ldots, y_n$. It is true if $L[y_i] = 0$ on $\mathcal{I}$ for each $i = 1, 2, \ldots, n$ (for some n^{th} order linear ordinary differential operator L).
- $\therefore$ To check for linear independence of solutions, it is enough to compute the value of the Wronskian at a single point in the interval $\mathcal{I}$.

Linear independence of solutions of linear ODEs

Example (Criterion for Linearly Independent Solutions)

Consider the ODE

$$y'' + y' - 2y = 0 \qquad (\heartsuit)$$

Two solutions are $y_1 = e^x$ and $y_2 = e^{-2x}$. *Check:*

$$y_1 = e^x \implies y_1' = e^x \implies y_1'' = e^x \implies y_1'' + y_1' - 2y_1 = 0$$
$$y_2 = e^{-2x} \implies y_2' = -2e^{-2x} \implies y_2'' = 4e^{-2x} \implies y_2'' + y_2' - 2y_2 = 0$$

Are y_1 and y_2 linearly independent? What is the general solution?

$$W(y_1, y_2) = W(e^x, e^{-2x}) = \begin{vmatrix} e^x & e^{-2x} \\ e^x & -2e^{-2x} \end{vmatrix} = -2e^{-x} - e^{-x}$$
$$= -3e^{-x} \neq 0 \text{ for any } x \in \mathbb{R}.$$

$\therefore \{y_1, y_2\}$ is a *fundamental set of solutions* of $(\heartsuit)$, and the *general solution* of $(\heartsuit)$ is $y = c_1 e^x + c_2 e^{-2x}$, where $c_1, c_2 \in \mathbb{R}$ are arbitrary.

Solutions of nonhomogeneous linear ODEs

Theorem (General Solution of Nonhomogeneous linear ODE [Trench 9.1.6])

*Suppose y_p is any particular solution of the nonhomogeneous n^{th} order linear ODE, $L[y] = g(x)$, on an interval $\mathcal{I}$, and suppose $\{y_1, y_2, \ldots, y_n\}$ is a fundamental set of solutions for the associated homogeneous equation, $L[y] = 0$, on $\mathcal{I}$. Then the **general solution** of $L[y] = g(x)$ on the interval $\mathcal{I}$ is*

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x) + y_p(x),$$

where c_i, $i = 1, 2, \ldots, n$, are arbitrary constants.

Note: $y_c(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x)$ is called the ***complementary function*** for the equation $L[y] = g(x)$. The general solution is $y = y_c + y_p$, where y_p is any solution of $L[y] = g(x)$.

Superposition of solutions of nonhomogeneous linear ODEs

Note: If L is a linear operator for which $L[y_1] = g_1(x)$ and $L[y_2] = g_2(x)$ then $L[y_1 + y_2] = L[y_1] + L[y_2] = g_1(x) + g_2(x)$, so:

Theorem (Superposition Principle for Nonhomogeneous Linear ODEs [Trench 9.1.7])

Suppose $y_{p_1}, \ldots, y_{p_k}$ are k particular solutions (on an interval $\mathcal{I}$) for the nonhomogeneous n^{th} order linear ODEs $L[y] = g_i(x)$, where $g_1(x)$, ..., $g_k(x)$ are continuous functions on $\mathcal{I}$. Then the superposition

$$y_p(x) = y_{p_1}(x) + y_{p_2}(x) + \cdots + y_{p_k}(x)$$

is a particular solution of

$$L[y] = g_1(x) + g_2(x) + \cdots + g_k(x).$$

Superposition of solutions of nonhomogeneous linear ODEs

Example (Nonhomogeneous superposition)

Consider the linear operator

$$L[y] = y'' + y' + y$$

Note that (writing $L[y] = y + y' + y''$ $\because$ it is more natural to compute)

$$L[x] = x + \frac{\mathrm{d}}{\mathrm{d}x}(x) + \frac{\mathrm{d}^2}{\mathrm{d}x^2}(x) = x + 1 + 0 = x + 1$$

$$L[x^2] = x^2 + \frac{\mathrm{d}}{\mathrm{d}x}(x^2) + \frac{\mathrm{d}^2}{\mathrm{d}x^2}(x^2) = x^2 + 2x + 2$$

$$L[x^3] = x^3 + \frac{\mathrm{d}}{\mathrm{d}x}(x^3) + \frac{\mathrm{d}^2}{\mathrm{d}x^2}(x^3) = x^3 + 3x^2 + 6x$$

Consequently, it follows from the superposition theorem that $y_p(x) = x + x^2 + x^3$ is a particular solution to

$$\begin{aligned} L[y] &= (x + 1) + (x^2 + 2x + 2) + (x^3 + 3x^2 + 6x) \\ &= x^3 + 4x^2 + 9x + 3 \end{aligned}$$

Practice problems: linear higher-order equations

In Trench, §9.1:

- Verifying a solution of an IVP: Exercise 1(a), (c)
- Checking a fundamental set: Exercise 6(a), (b)
- Computing Wronskians: Exercise 10(a)–(c)
- Uniqueness and the zero solution: Exercise 11
- Linear dependence of functions: Exercise 16